AUTOMATED DAILY LITERATURE DIGEST • PhD RESEARCH WATCH

Particulate Matter
Monitoring & Health Effects

Issue 2026-11-05 | Entry window 5 November 2026 (full day) | 25 records screened in scope

25 RECORDS IN SCOPE (63 EXCLUDED)

9 OF 10 SUBTOPIC BANDS POPULATED

6 EFFECT ESTIMATES WITH A CI

2 TIER-A RECORDS

5 RECORDS CARRIED ON METADATA ALONE

Signal of the day

A dense low-cost $PM_{2.5}$ network, gap-filled by machine learning, was carried all the way to district-level short-term mortality across Bihar in *Nature Communications*: northern districts ran 1.2 times the mean exposure of southern ones, district mortality varied 1.7-fold and daily attributable deaths 3.2-fold.

IIT Kanpur (Tripathi group); network $PM_{2.5}$ fused with NFHS socioeconomic data and India-specific relative risks. State-level exposure inequity was limited; inter- and intra-regional inequity within Bihar was substantial. **Where it is weak:** the abstract reports no node count, no calibration R^2 or RMSE and no uncertainty on attributable deaths. The mortality is a risk function applied to a sensor-derived surface, not an observed association, so any network calibration bias passes one-for-one into the burden. **Why it matters here:** this is the clearest high-visibility case of low-cost sensing as the exposure layer of a burden estimate outside cities. Its value to the field rests on the calibration and propagated-uncertainty detail in the full text, which the weekly should pull.

What changed today

- **Sinonasal cancer, and a window gradient.** US commercial claims, 3,600 cases and 17,695 controls: per SD of $PM_{2.5}$ (about 2.5 $\mu g/m^3$), aOR 1.25 (1.05–1.49) at 6 months, 1.31 (1.08–1.59) at 1 year, 1.36 (1.11–1.66) at 5 years. Exposure is a 1-km satellite-fused product collapsed to metropolitan area, case-control contrast is 0.2–0.3 $\mu g/m^3$, and wood and leather dust, the main SNM cause, is unmeasured.
- **Two influenza designs, one PM magnitude.** Shanxi 11-city time-series: RR 1.024 (1.015–1.034) per 10 $\mu g/m^3$ $PM_{2.5}$, but NO_2 (1.115) and SO_2 drive the mixture. Shanghai laboratory-confirmed case-crossover (BACKFILL): PM_{10} OR 1.019 (1.004–1.034). Both near 2 % per 10 $\mu g/m^3$, both on regulatory monitors; gases carry more signal than PM.
- **Where the dose is, mass is not.** Chamber emission of six indoor sources: most mass and the highest mass-normalised DTT OP sit below 0.25 μm, and for equal PM_{10} mass these sources give an estimated 76 ± 10 % more alveolar dose than ambient PM. Kanpur–Mumbai impactor pairs: fine mode delivers 53–75 % of alveolar dose from 40–62 % of mass. Both point at the size range optical sensors count worst.
- **Composition breaks transfer.** Around an opencast coal mine (BACKFILL): roadside $PM_{2.5}$ near 400 $\mu g/m^3$, Cr, Ni, Fe, Zn 5–10-fold above background, OP only weakly explained (25–40 %). Shenzhen fireworks: 4.7 % of $PM_{2.5}$ over the season but >50 % at peak hours. Accra FRM series (BACKFILL): three sites at 3–4 times the WHO guideline, a reference anchor for West African sensor calibration.
- **Exposure basis (x1).** Of 25 records, 1 rests on a low-cost network (Bihar, on the sensing side of x1) and none on a personal monitor. Health side by subtopic (n=7 with a recoverable basis): 71 % measured, 14 % satellite-fused, 14 % proxy — but the measured share is three regulatory-monitor respiratory studies and two occupational filter campaigns. *All six effect estimates below come from regulatory monitors or an MSA-averaged satellite product*; none is tied to a sensor-measured exposure.

Window continuity

The 4 November issue closed with `last_entry_date` at 4 November. This issue was built on the **6 November 17:34 UTC (11:34 CST) run**, the 05:04 UTC slot firing 12.5 hours late. Before 22:00 the newest buildable day is yesterday, so **5 November is covered in full**. Thursday: no rollup is owed; **W45 (1–7 November) falls on Saturday 7 November**. Three BACKFILL records, the cap: two held from the 4 November run (Accra FRM, Shanghai influenza) and one from Amass (coal-mine $PM_{2.5}$ composition). **Europe PMC returned zero records for the day** (likely indexing lag; the leg is not broken on 4 Nov). The OpenAlex connector was again not available on this session.

Corpus analytics

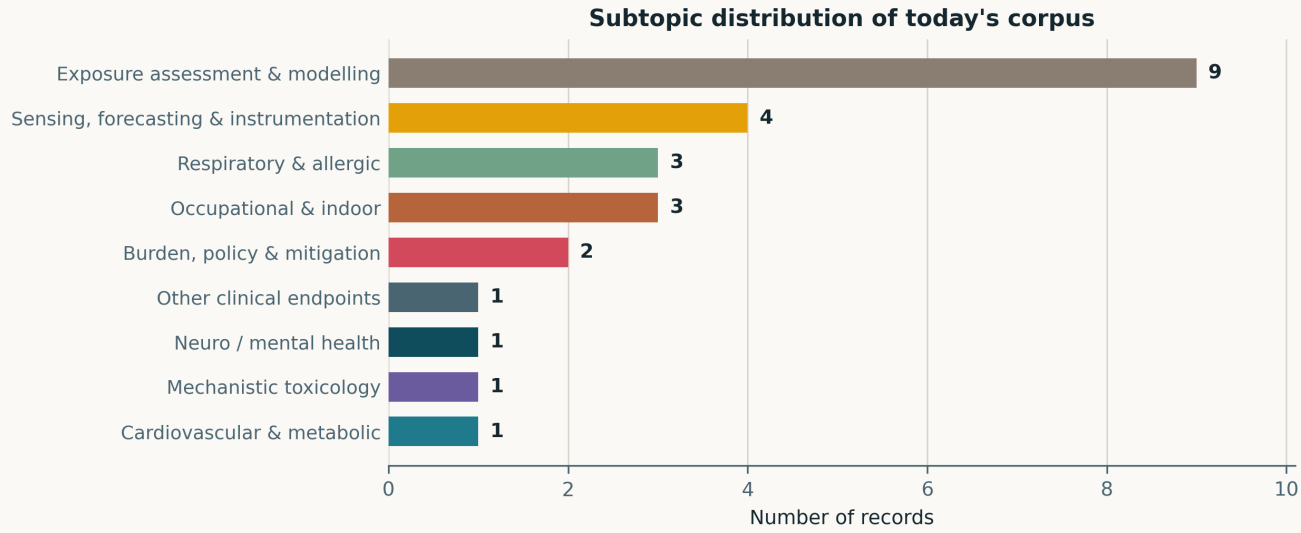


Fig. Subtopic assignment of the 25 records across nine of ten bands: exposure assessment 9, sensing 4, respiratory 3, occupational and indoor 3, burden 2, and one each for other clinical, neuro, mechanistic and cardiovascular. Reproductive is empty today.

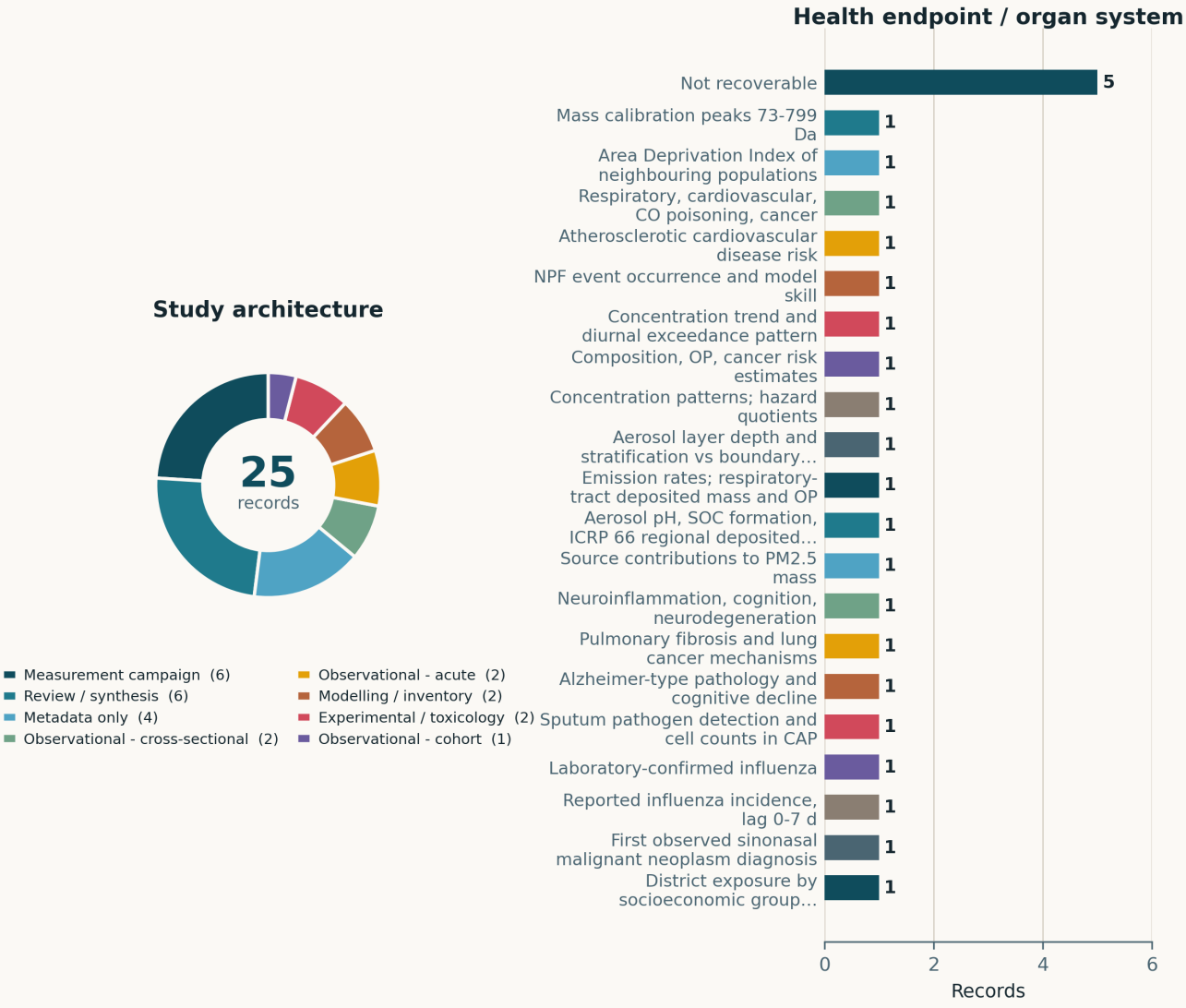


Fig. Left — architecture: measurement campaign 6, review or synthesis 6, metadata-only 4, and 2 each cross-sectional, acute observational, modelling and experimental; one cohort (CAPNETZ). Right — endpoints are singletons apart from 5 unrecoverable (four Elsevier metadata-only records and the UFP letter); nine of the twenty labels are measurement or chemistry outcomes, not clinical ones.

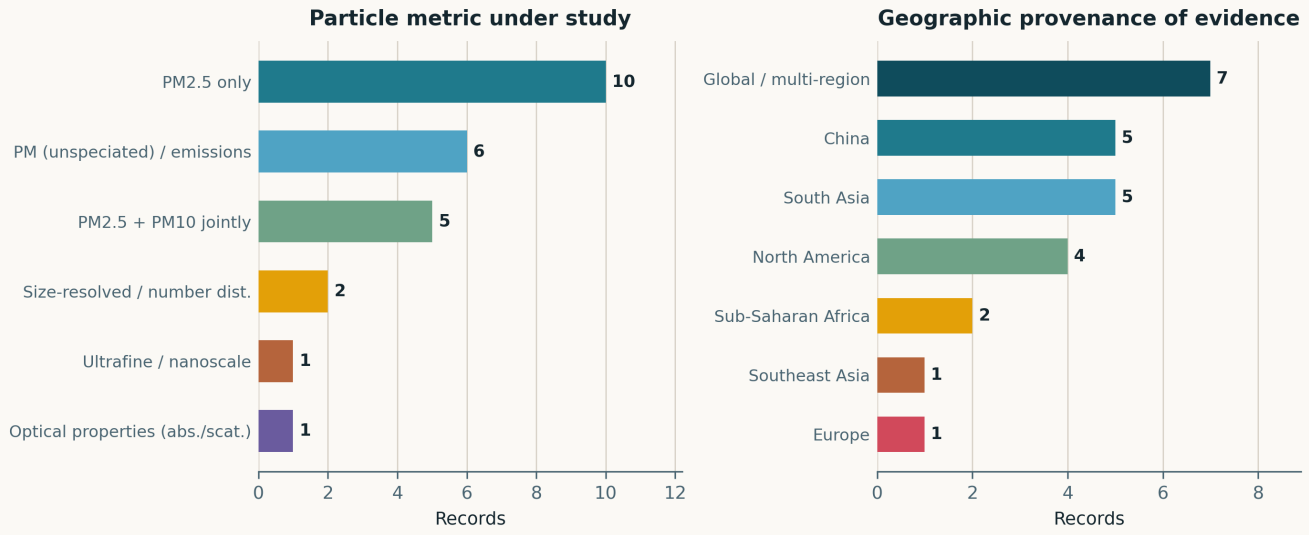


Fig. Left — particle metric: $PM_{2.5}$ only 10, unspeciated or emissions 6, $PM_{2.5}$+PM_{10} jointly 5, size-resolved 2, optical 1, ultrafine 1. Right — geography: global or multi-region 7 (four reviews, the UFP letter, one metadata-only record with no location and the laboratory AMS study, which fall there by default), South Asia 5 (India 4, Pakistan 1 counting the regional review), China 5, North America 4, sub-Saharan Africa 2, and one each Europe and Southeast Asia.

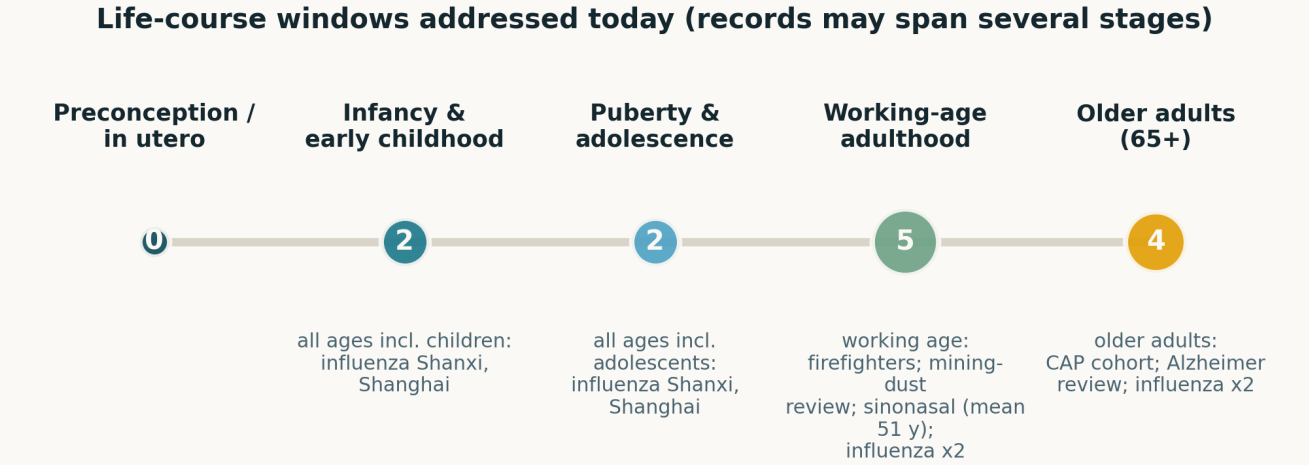


Fig. Life-course windows. The two influenza series are all-age and are counted in every postnatal stage; working age adds firefighters, the dust review and the sinonasal case-control (case mean age 51); older adults add the CAP cohort and the Alzheimer review. No preconception or in-utero record today.

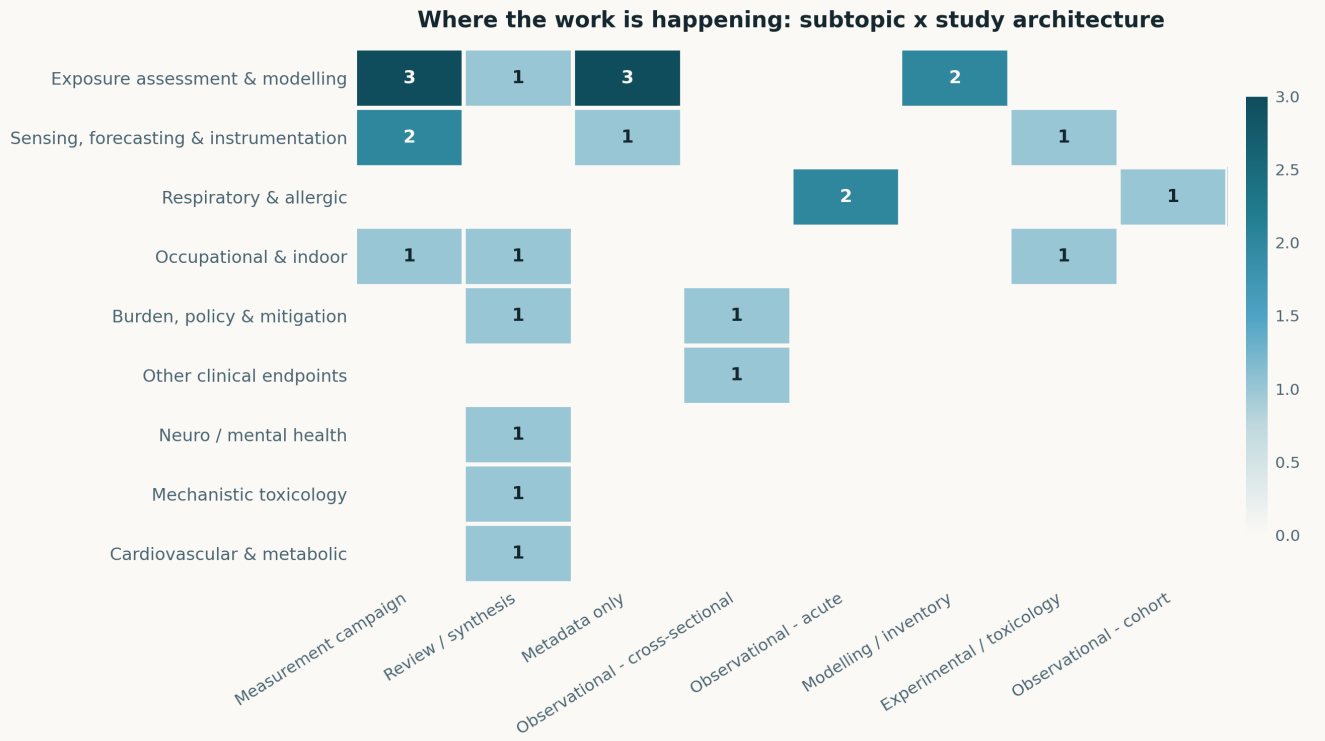


Fig. Subtopic × study architecture. Exposure assessment splits across measurement (3), metadata-only (3), modelling (2) and the letter; sensing is two measurement records, one metadata-only and the AMS calibration. The only cohort is respiratory (CAPNETZ), and neither respiratory acute design uses a sensor.

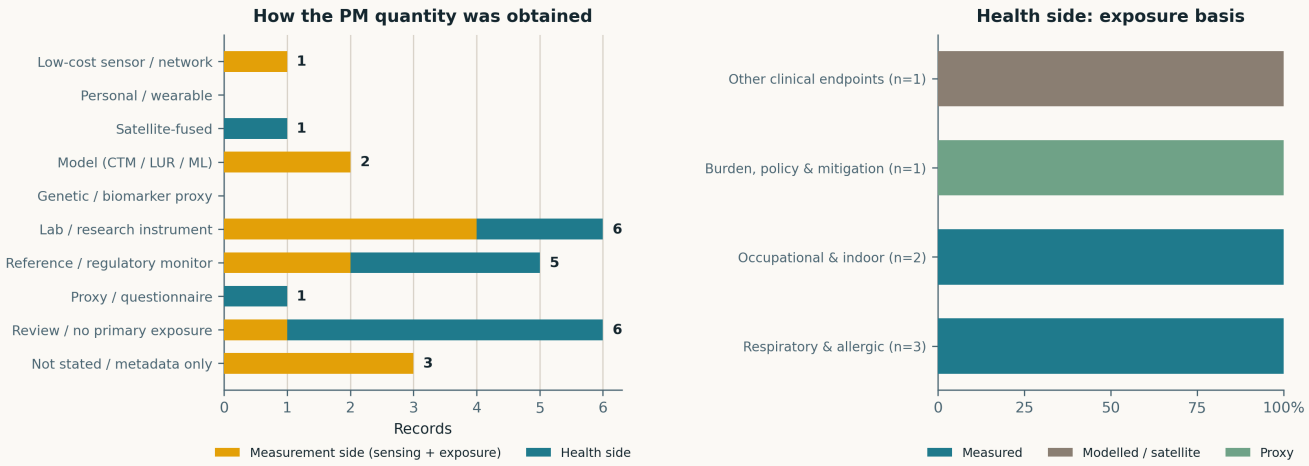


Fig. How each record obtained its PM quantity, classified from the exposure field. 1 of 25 records rest on a low-cost sensor or sensor network; 5 on reference or regulatory monitors. Health side, where an exposure basis is recoverable (n=7): measured 71 %, modelled or satellite-fused 14 %, proxy 14 %. Right panel sorted by modelled share: the subtopics at the bottom are where a measured low-cost network exposure would add most.

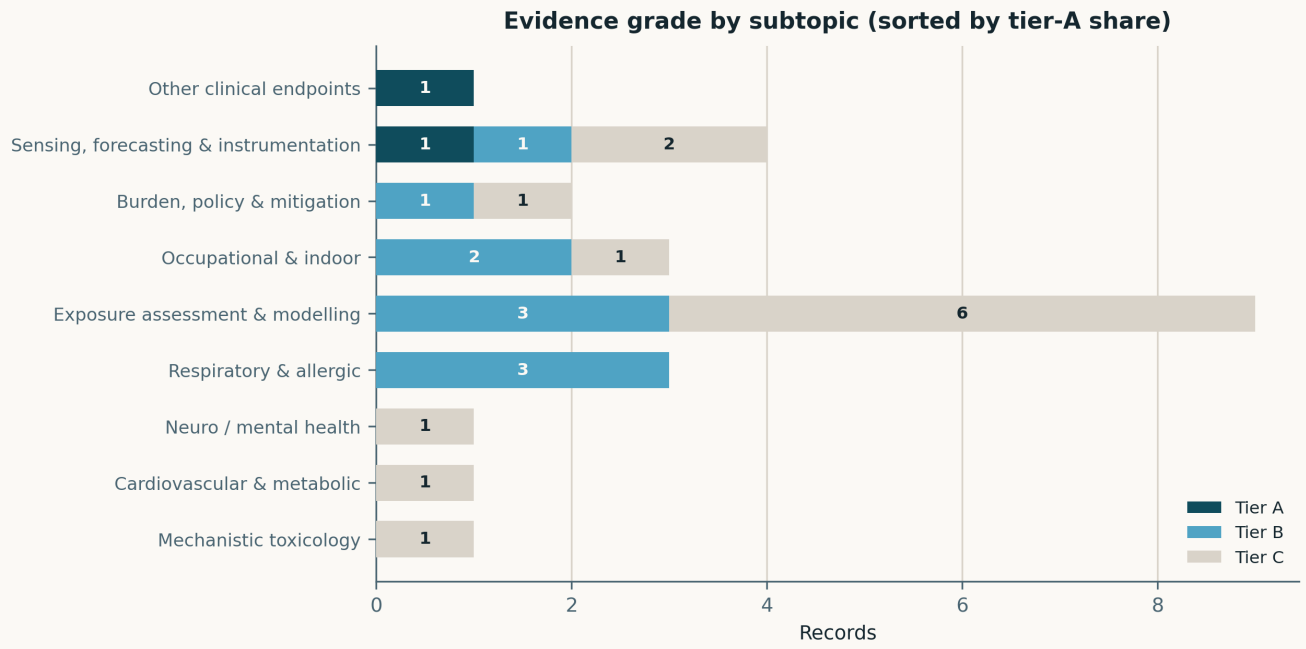


Fig. Tier mix per subtopic. 2 of 25 records are tier A (8 %). Highest tier-A share: *Other clinical endpoints* (1 of 1).

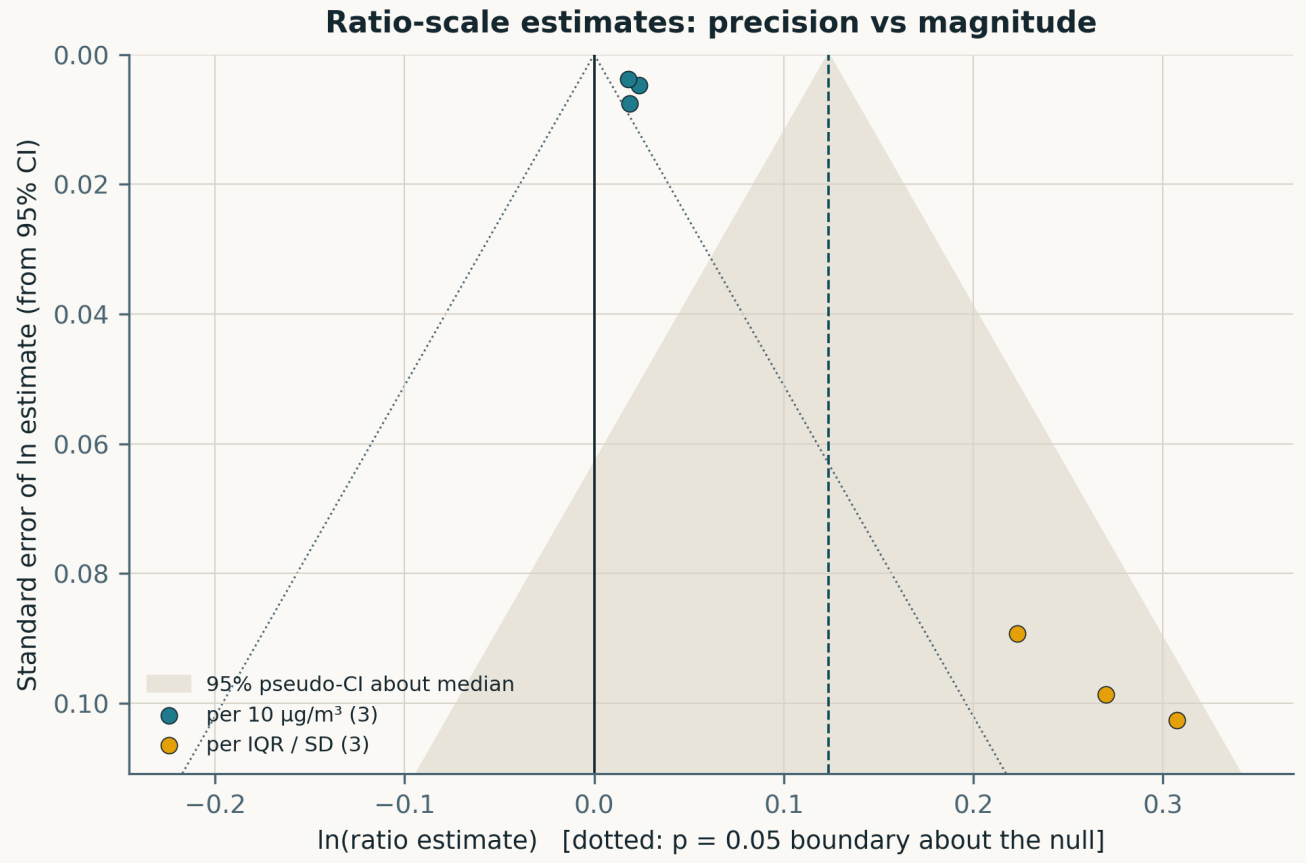


Fig. Every ratio-scale estimate (n=6) plotted as ln(estimate) against its standard error, coloured by the exposure increment the authors reported. 6 of 6 exclude the null. Increments: per IQR / SD 3, per 10 $\mu g/m^3$ 3. Because increments differ (per 1 unit, per 10 $\mu g/m^3$, per IQR, categorical), the spread is not a heterogeneity estimate and the median line is not a pooled effect; the plot shows whether precision and magnitude co-vary.

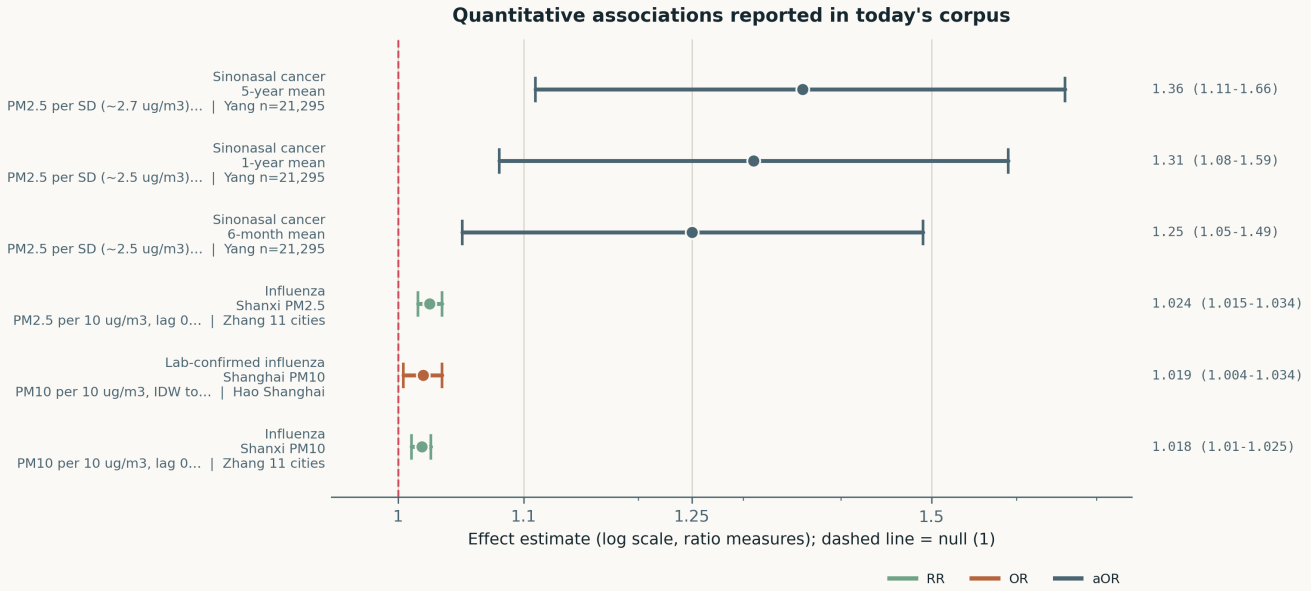


Fig. Ratio scale, 6 estimates from 3 sources: Yang sinonasal case-control (3, per SD), Zhang Shanxi influenza (2, per 10 $\mu g/m^3$), Hao Shanghai influenza (1, per 10 $\mu g/m^3$). The three sinonasal estimates are nested averaging windows of one exposure in one sample and are not independent. The three influenza estimates share endpoint family and increment, but two are PM_{10} and one $PM_{2.5}$, and RR from a time-series is not OR from a case-crossover; no pooling and no I^2.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase. Records marked *metadata only* carry no recoverable abstract and assert no finding.

Neuro / mental health — 1 record

Defrancesco lung-brain axis review $PM_{2.5}$ Alzheimer disease mechanisms 2026 Clin Sci (Lond)

Mechanistic understanding of the role of the lung-brain axis in the pathogenesis of $PM_{2.5}$-induced Alzheimer's disease.

Narrative review positioning pulmonary injury, rather than direct translocation, as the main route from inhaled $PM_{2.5}$ to AD pathology: systemic inflammation, oxidative stress, blood-brain barrier disruption and extracellular-vesicle signalling converge on amyloid, tau, microglial and autophagy-lysosomal dysfunction. Modifiers named are composition, co-pollutants, sex, APOE and epigenetic reprogramming. No search protocol, no quantitative synthesis; the framework is a hypothesis map and the human causal link remains epidemiological association. Relevance: indirect - if the pulmonary route dominates, deposited dose in the alveolar region, not ambient mass, is the exposure metric a sensor should be calibrated to serve.

doi: 10.1042/cs20262109

Cardiovascular & metabolic — 1 record

Siddiqui narrative review AQI ASCVD South Asians 2026 Curr Atheroscler Rep

Impact of Air Quality Index on Atherosclerotic Cardiovascular Disease Risk in South Asians: A Narrative Review.

Narrative review arguing that worse AQI and its constituent pollutants raise ASCVD risk in South Asians, who already carry high genetic and metabolic predisposition, and that pollution may compound traditional risk factors. Reports no search strategy, no pooled estimate and no South Asia-specific effect sizes; it concedes that region-specific intervention evidence is thin. Using AQI as the exposure is the core weakness: it is a max-of-pollutants communication index, not a dose metric, and conflates PM with gases. Relevance: indirect - the region with the highest burden still lacks cohort-grade exposure, which is where dense sensing would add most.

doi: 10.1007/s11883-026-01486-0

Respiratory & allergic — 3 records

Zhang Shanxi 11-city time-series pollutant mixtures influenza vaccination 2026 Environ Res

Influenza Incidence in Relation to Air Pollution Mixtures and the Modifying Role of Vaccination in Shanxi Province, China.

11 Shanxi cities, 2014-2022, DLNM with zero-inflated negative binomial and meta-analysis. Per 10 $\mu g\,m^{-3}$ $PM_{2.5}$, RR 1.024 (1.015-1.034); PM_{10} 1.018 (1.010-1.025); NO_2 1.115 (1.086-1.146). WQS (1.078, 1.005-1.157) and qgcomp (1.122, 1.064-1.182) mixture effects were driven by SO_2 and NO_2, not PM. Vaccination modification is city-level coverage, therefore ecological, and labelled exploratory. Surveillance incidence is care-seeking and testing dependent; 2020-2022 includes COVID-era behavioural shifts the abstract does not say were handled. Relevance: indirect - PM signal is small next to gases, which argues for multi-pollutant nodes rather than PM-only networks for infectious endpoints.

doi: 10.1016/j.envres.2026.126109

Hao Shanghai case-crossover lab-confirmed influenza six pollutants 2026 Int J Biometeorol

Association between short-term air pollutant exposure and influenza incidence in Shanghai, China: A case-crossover study

Laboratory-confirmed influenza cases, Shanghai 2013-2017, time-stratified case-crossover with IDW exposure at the individual level. Per 10 $\mu g\,m^{-3}$ PM_{10}, OR 1.019 (1.004-1.034); O_3 was inverse (OR 0.959, 0.941-0.977). $PM_{2.5}$, SO_2, NO_2 and CO were positive across lags but their point estimates are not in the abstract. Laboratory confirmation is the design's strength over syndromic counts; IDW from a sparse regulatory grid and selection into testing are the weaknesses, and the negative O_3 term likely absorbs season. Read alongside the same-day Shanxi series: two independent Chinese designs give PM effects near 2% per 10 $\mu g\,m^{-3}$. Relevance: indirect. BACKFILL, entry 20 March 2026.

doi: 10.1007/s00484-026-03163-y

Kutzinski CAPNETZ air pollution sputum pathogen spectrum community-acquired pneumonia 2026 Pneumonia (Nathan)

Air pollution is associated with a distinct pathogen spectrum in community-acquired pneumonia: a prospective multicentre cohort study.

CAPNETZ, 4,446 CAP patients with sputum; pathogen found in 1,581, cell counts in 4,014. Mixed-effects logistic models: S. pneumoniae with SO_2 (OR 1.75, p=0.008) and CO (1.57); E. coli with NO_2 (2.75); higher $PM_{2.5}$ with lower detection of the pathogen examined (OR 0.37, p=0.001). No CIs or increments in the abstract, so magnitudes are ungradable. City-level exposure over a month is coarse, many pollutant-pathogen pairs were tested, and an inverse $PM_{2.5}$ association invites a sampling or antibiotic-pretreatment explanation the abstract does not exclude. Relevance: indirect - outcome definition, not exposure metrology, limits this.

doi: 10.1186/s41479-026-00212-3

Mechanistic toxicology — 1 record

Sun Waite review molecular mechanisms respirable silica coal dust asbestos lung disease 2026 Part Fibre Toxicol

Molecular mechanisms of dust-induced lung disease: exploring the role of physicochemical composition and cellular pathways.

Review of respirable occupational dust toxicity: oxidative stress, mitochondrial and lysosomal damage, inflammasome activation, ferroptosis, necrosis and pyroptosis, macrophage/neutrophil activation and NK dysfunction driving fibrosis and tumorigenesis, with emphasis on silica and on coal dust bearing reduced iron. Covers antifibrotic and immunotherapy approaches. Narrative, no stated search or appraisal. The physicochemical angle (surface iron, crystallinity) is the useful part: it is why mass concentration alone mis-ranks hazard across dust types. Relevance: indirect - optical low-cost sensors are blind to mineralogy, so occupational dust monitoring needs composition or a source-specific mass-to-hazard factor.

doi: 10.1186/s12989-026-00696-3

Exposure assessment & modelling 9 records

Tan Shenzhen PMF-ML firework burning $PM_{2.5}$ source apportionment 2026 Environ Res

Apportioning Firework Burning and Routine Source Contributions to PM(2.5) in a Chinese Megacity Using Two PMF-Machine Learning Frameworks.

Three Spring Festival seasons of hourly speciation in Shenzhen; PMF resolved 11 routine sources plus firework burning. Fireworks contributed 4.7% of $PM_{2.5}$ over the full period, 33% during display episodes and over 50% at peak hours. Two ML surrogates (direct and indirect) reproduced PMF contributions and flagged fireworks as dominant during episodes. Agreement with PMF is circular validation - the ML learns PMF output, so it inherits PMF's factor ambiguity and no independent tracer test is reported. Relevance: direct for networks - episodic sources at >50% of hourly mass are exactly where a fixed humidity/composition calibration on optical sensors breaks.

doi: 10.1016/j.envres.2026.126121

Sathish Kanpur Mumbai nine-stage size-resolved aerosol chemistry ICRP deposited dose 2026 Environ Res

Size-resolved measurements matter for chemistry and health: paired inland-coastal size-resolved aerosol chemistry and the inadequacy of bulk PM(2.5) dose estimates.

Paired Kanpur (inland IGP) and Mumbai (coastal) impactor sampling over three seasons. Fine mode (PM2.1) is up to 62% of mass in Kanpur, not Mumbai; Kanpur fine aerosol is more acidic (pH 3.5-4.5 vs 4.5-5.0). ICRP 66 dosimetry: fine mode delivers 53-75% of alveolar dose from 40-62% of mass, and Kanpur alveolar doses are 1.3-2 times Mumbai's. The authors flag that SOC-water associations lose significance at one value per sample day, and dose depends on assumed deposition fractions. Relevance: direct - a calibration fitted at a coastal site will misweight the inland fine mode that carries the dose.

doi: 10.1016/j.envres.2026.126094

Ayittey Accra FRM gravimetric $PM_{2.5}$ three sites e-waste 2021-2022 2026 Environ Monit Assess

Persistent urban $PM_{2.5}$ pollution in Accra, Ghana: spatiotemporal patterns, meteorological and anthropogenic drivers, and associated health risks

Two years (2021-2022) of FRM gravimetric $PM_{2.5}$ at Adabraka (next to the Agbogbloshie e-waste burning area), Dansoman and Legon. All sites ran about 3-4 times the WHO 24-h guideline; Adabraka was highest at 60.76 (2021) and 60.54 $\mu g\,m^{-3}$ (2022), with negligible interannual change but strong spatial contrast. PCA ties variability positively to temperature and negatively to RH. Hazard quotients exceeded 1 for adults and children. Sampling was intermittent (~300 samples over 3 sites and 2 years), and HQ applied to PM mass is a screening device, not an epidemiological risk. Relevance: direct - one of very few reference-grade West African series, usable to anchor low-cost and satellite calibration. BACKFILL, entry 3 March 2026.

doi: 10.1007/s10661-026-15117-3

Jaelani Surabaya Benowo landfill 15-min $PM_{2.5}$ 2020-2024 2026 Environ Sci: Atmos

The spatial impact assessment of landfill sites on air quality in Surabaya: a diurnal and seasonal perspective

Only a one-sentence abstract is available. Five years of 15-minute monitoring at the Benowo landfill show $PM_{2.5}$ rising from 37.17 to 50.32 $\mu g\,m^{-3}$ between 2020 and 2024, 7-10 times the WHO annual guideline, with exceedance peaks at 07:00 and overnight. The authors attribute the timing to boundary-layer mixing rather than landfill operations. Instrument class, completeness and any upwind comparison are not stated, so landfill attribution cannot be judged. Relevance: indirect - the diurnal peak set by mixing depth is a reminder that a single fixed node cannot separate source from meteorology without a paired background site.

doi: 10.1039/d6ea00048g

Ku North China Plain new particle formation GEOS-Chem-APM spring campaign 2026 Environ Sci Technol

Investigations of New Particle Formation Mechanisms in North China Plain during the Spring Campaign Based on Observations and Simulations

One-month 2016 spring campaign in an industrial North China Plain city: 10 NPF days, each after nighttime northwesterly ventilation under dry daytime conditions. GEOS-Chem-APM reproduced 6 of 10 events; NH_3 was in excess; tripling H2SO4 raised nucleation but did not reproduce particle evolution, and simulated growth rates were 51% and 75% too low on well- and poorly-simulated days. Ten events and one season limit generality, and the campaign is a decade old. No mass or health metric. Relevance: indirect - NPF bursts raise ultrafine number with little $PM_{2.5}$ mass, the regime mass-calibrated optical nodes cannot see.

doi: 10.1021/acs.est.6c07540

Schwarz Helmholtz letter ultrafine particle health studies forest for the trees 2026 Environ Res

Letter to the Editor: Health effect studies of ultrafine air pollution-Are we missing the forest for the trees?

Metadata only; a letter with no abstract, so no argument is asserted beyond the title. From the Helmholtz Munich epidemiology group (Schwarz, Pickford, Cyrys, Breitner-Busch), which runs long-term UFP number and size measurement in Augsburg. The title signals a critique of how UFP health studies are framed - plausibly metric choice (number vs surface vs size bins) or the pursuit of UFP-specific effects independent of correlated traffic pollutants. To be read when the full text is accessible. Relevance: possible - any UFP exposure standard decides what number-concentration sensors in a network should report.

doi: 10.1016/j.envres.2026.126091

Hussain Bergin Pakistan smog episodes data-driven source attribution 2026 Atmos Environ

Attribution of Air Pollution Sources During Smog Episodes Across Major Urban Centers of Pakistan: A Data-Driven Approach

METADATA ONLY. Metadata only; no abstract in Crossref at issue time, so no finding is asserted. The title promises data-driven source attribution for smog episodes across several Pakistani cities; the author list includes Mike Bergin, whose group has built low-cost and satellite $PM_{2.5}$ work in South Asia, and Muhammad Fahim Khokhar. Carried because Punjab smog is among the most extreme PM regimes, and attribution there decides between crop burning, brick kilns and traffic. To be assessed when the abstract deposits: data sources (sensor, reference, satellite) and method. Relevance: possible, unassessed.

doi: 10.1016/j.atmosenv.2026.122473

Shankar meteorological drivers urban air quality weather-pollution interactions 2026 Atmos Environ

Understanding the Meteorological Drivers of Urban Air Quality: Weather-Air Pollution Interactions

METADATA ONLY. Metadata only; no abstract in Crossref at issue time, so no finding is asserted. The title points to an analysis, or possibly a review, of how weather variables drive urban air quality. Carried provisionally: meteorological normalisation is a prerequisite for interpreting network trends and for separating calibration drift from real change. Geography, pollutants and design are unknown, and it may be gas-dominated, in which case it will be noted as such in the weekly. Relevance: possible, unassessed.

doi: 10.1016/j.atmosenv.2026.122468

Yao Harbin early-spring heavy pollution episodes natural anthropogenic drivers 2026 Atmos Pollut Res

Natural and anthropogenic drivers of heavy pollution episodes in early spring in Harbin, Northeast China

METADATA ONLY. Metadata only; no abstract in Crossref at issue time, so no finding is asserted. The title sets natural against anthropogenic drivers of early-spring heavy pollution in Harbin, a period where heating-season combustion, agricultural residue burning and stagnant boundary layers overlap. Carried because cold, humid episodes are the regime where optical low-cost sensors overestimate most and where humidity correction matters. To be assessed when the abstract deposits: whether it separates open burning from heating and uses PM composition. Relevance: possible, unassessed.

doi: 10.1016/j.apr.2026.103263

Sensing, forecasting & instrumentation — 4 records

Agrawal Bihar low-cost sensor network ML exposure inequity short-term mortality 2026 Nat Commun

Unveiling hidden air pollution exposure and impacts with low-cost sensor network-based frameworks.

Dense low-cost $PM_{2.5}$ network across Bihar, fused with ML surfaces, NFHS socioeconomic data and India-specific relative risks. Northern districts carried 1.2 times the mean exposure of southern ones; district mortality per 100,000 varied up to 1.7-fold and daily attributable mortality 3.2-fold. State-level exposure inequity was limited, intra-regional inequity substantial. The abstract gives no sensor count, calibration R2/RMSE, or uncertainty on the attributable deaths, and the mortality is borrowed risk functions applied to sensor-derived surfaces, not an observed association; network calibration error propagates straight into the burden. Relevance: direct - the clearest flagship case for low-cost networks as a burden-estimation input outside cities.

doi: 10.1038/s41467-026-77148-1

Chen TRACER Houston micropulse lidar aerosol vertical profile boundary layer parameterisation 2026 Atmos Chem Phys

Aerosol vertical distributions shaped by boundary layer dynamics in a coastal urban environment: insights from the TRACER campaign

Six months of TRACER (Houston, 2022) micropulse lidar retrievals anchored to ground aerosol data. An S-shaped fitting function yields an aerosol-layer depth and a stratification-strength parameter. Concentrations drop sharply at strong capping inversions, layer depth tracks boundary-layer growth and collapse, and sea-breeze passage swaps continental for maritime aerosol. Relationships with BLH and the potential-temperature gradient are statistically significant but weak (no r given). No health endpoint. Relevance: direct for satellite-surface fusion - the AOD-to-surface $PM_{2.5}$ ratio depends on exactly this profile shape, and the parameterisation offers a physical predictor in place of BLH alone.

doi: 10.5194/acp-26-15543-2026

Wang $PM_{2.5}$ PM_{10} forecasting dual-data MWS-optimised ensemble learning 2026 Atmos Environ

$PM_{2.5}$ and PM_{10} Concentration Forecasting Using a Dual-Data Strategy and MWS-Optimized Ensemble Learning

METADATA ONLY. Metadata only; no abstract in Crossref at issue time, so no finding is asserted. The title describes $PM_{2.5}$ and PM_{10} forecasting from a dual-data input strategy with an ensemble learner tuned by an MWS optimiser. Carried because short-horizon forecasting is the step most networks attempt after calibration, and ensemble skill claims usually hinge on whether validation is time-blocked. To be assessed when the Elsevier abstract deposits: horizon, held-out period, skill against persistence, and whether sensor or reference data feed it. Relevance: possible, unassessed.

doi: 10.1016/j.atmosenv.2026.122424

Rogers VUV-AMS siloxane radical mass calibration source aerosol mass spectrometry 2026 J Phys Chem A

Siloxane-Radical Species: Chemical Characteristics and Use for Mass Calibration in Aerosol Mass Spectrometry.

Synchrotron VUV photoionisation aerosol mass spectrometry: heating conductive silicone tubing (150-230 C) inline with a soot source deposits siloxane oligomers on particles, producing three substituted-siloxane radical series and 21 recurring peaks between 73 and 799 Da. Calibration was verified on five PAHs (178-300 Da), 2,5-dimethylfuran and ethylene-pyrolysis hydrocarbons (165-374 Da). The radical-formation mechanism on the CuO target is hypothesised, not shown, and the method is specific to one research-grade platform. Relevance: indirect - siloxane from silicone sample lines is also a known artefact; the paper doubles as a warning for any sensor or sampler plumbed with heated silicone tubing.

doi: 10.1021/acs.jpca.6c04791

Occupational & indoor — 3 records

Kumar firefighter occupational exposure neurotoxicity narrative review 2026 Toxicol Mech Methods

Occupational exposure and neurotoxic health effects in firefighters: toxicological mechanisms and health risks.

PRISMA-informed narrative review of literature to June 2026, without meta-analysis or risk-of-bias assessment. Proposed pathways: CO and HCN hypoxia with ATP depletion acutely; fine PM and PAHs reaching the brain, NLRP3 glial activation and ferroptosis chronically; heat, fatigue and sleep loss as amplifiers. The authors state the firefighter literature is mostly cross-sectional and the mechanisms come from non-firefighter models, so causality is uncertain. Relevance: indirect - fireground PM exposure is essentially unmeasured at the individual level; wearable PM sensing on turnout gear is the obvious gap.

doi: 10.1080/15376516.2026.2748390

Subramanian chamber size-resolved emission rates PM oxidative potential six indoor sources 2026 Environ Sci Technol

Size- and Time-Resolved Emission Rates of Particulate Matter Mass, Chemical Constituents, and Oxidative Potential from Indoor Sources

Chamber tests of two incenses, candle, toaster, humidifier and cigarette. Emission was mostly unimodal and concentrated below 0.25 um, where mass-normalised DTT OP was highest; composition and OPm varied with size but little with time. 37 +/- 3% of emitted PM_{10} mass and OP is deposition-capable, 45 +/- 2% of it alveolar. For equal PM_{10} mass and OP, these indoor sources give an estimated 76 +/- 10% higher alveolar dose than ambient PM. Six sources in one chamber; the dose comparison is modelled, not measured. Relevance: direct - optical sensors undercount below ~0.3 um, so indoor nodes systematically miss the fraction carrying most OP.

doi: 10.1021/acs.est.6c08882

Gupta Maharashtra opencast coal mine $PM_{2.5}$ composition oxidative potential 2026 Environ Geochem Health

Comprehensive assessment of $PM_{2.5}$ exposures around a regional mining site in India: integrating chemical composition, oxidative potential and health risk

Six sites around an active opencast coal mine in eastern Maharashtra: two roadside (~400 $\mu g\,m^{-3}$), residential at 2 and 5 km (122 +/- 43 and 116 +/- 36 $\mu g\,m^{-3}$), and background. Carbonaceous species and metals were ~40-50% of mass; Cr, Ni, Fe, Mg and Zn ran 5-10-fold higher near the mine. OPvAA and OPvDTT peaked roadside and correlated only weakly with metals and organics (r=0.3-0.5); regression explained 25-40% of OP. EC cancer risk ~1e-3 rests on a risk-assessment slope, not outcomes. Season and sample counts are not in the abstract. Relevance: direct - at this composition, optical-sensor mass calibrations from urban sites will not transfer. BACKFILL, entry 25 June 2026.

doi: 10.1007/s10653-026-03301-w

Burden, policy & mitigation — 2 records

Laban scoping review household backup generators emissions health Africa 2026 BMC Public Health

Health effects associated with emissions from household generators used for backup power in Africa: a scoping review.

PubMed and Google Scholar scoping review; nine studies met inclusion, and only ~2% of screened studies addressed health outcomes directly. Generator emissions carried $PM_{2.5}$, PM_{10}, NOx and SO_2 at high levels; reported links to respiratory symptoms, CO poisoning, and possibly cancer and hypertension rest on self-report, modelling or indirect evidence. No study established a causal link. Two databases, English only. Relevance: direct - generator plumes are intermittent, near-field and household-specific, a source that only distributed indoor/outdoor low-cost sensing can capture at the needed resolution in Nigeria, Ghana or South Africa.

doi: 10.1186/s12889-026-28946-1

US coal-fired power plant proximity Area Deprivation Index cross-sectional 2026 JAMA Netw Open

Distribution of Coal Power, Pollution, and Poverty.

2023 Neighborhood Atlas and census block groups: 2,783,004 people (0.83% of the US) live within 3 miles of a coal plant, disproportionately in the most deprived ADI decile and more often American Indian/Alaska Native. The most advantaged decile had OR 0.04 (0.00-0.81) of living within 3 miles versus the most deprived - a CI spanning near-zero that signals sparse cells. Proximity stands in for exposure; no PM, metal or stack-emission data, and plant size and wind direction are ignored. Relevance: indirect - a siting map for fenceline nodes, not an exposure estimate.

doi: 10.1001/jamanetworkopen.2026.42335

Other clinical endpoints — 1 record

Yang US claims case-control $PM_{2.5}$ sinonasal malignant neoplasm 2026 JAMA Otolaryngol Head Neck Surg

Fine Particulate Matter Exposure and Sinonasal Malignant Neoplasm.

National commercial-claims case-control, 2007-2020: 3,600 SNM cases, 17,695 controls (~1:5). Per SD $PM_{2.5}$ (about 2.5 $\mu g\,m^{-3}$), adjusted OR 1.25 (1.05-1.49) for the 6-month mean, 1.31 (1.08-1.59) for 1 year and 1.36 (1.11-1.66) for 5 years, rising with averaging window. Exposure is assigned at metropolitan-area level, so a 1-km product is collapsed to an ecological contrast of ~0.2-0.3 $\mu g\,m^{-3}$ between groups; smoking and deprivation are regional, not individual, and occupational wood/leather dust - the dominant SNM cause - is unmeasured. Cases are 10 years older than controls. Relevance: indirect - a rare endpoint where residence-level exposure from dense sensing would remove the MSA misclassification.

doi: 10.1001/jamaoto.2026.3453

Corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Defrancesco lung-brain axis review $PM_{2.5}$ Alzheimer disease mechanisms	Mechanistic understanding of the role of the lung-brain axis in the pathogenesis of $PM_{2.5}$-induced Alzheimer's disease.	*Clin Sci (Lond)*	Mechanistic review	$PM_{2.5}$ (general; composition discussed as modifier)	Global
Cardiovascular & metabolic					
Siddiqui narrative review AQI ASCVD South Asians	Impact of Air Quality Index on Atherosclerotic Cardiovascular Disease Risk in South Asians: A Narrative Review.	*Curr Atheroscler Rep*	Narrative review	AQI and constituent pollutants ($PM_{2.5}$, PM_{10}, NO2, CO, O3, NO)	South Asia
Respiratory & allergic					
Hao Shanghai case-crossover lab-confirmed influenza six pollutants	Association between short-term air pollutant exposure and influenza incidence in Shanghai, China: A case-crossover study	*Int J Biometeorol*	Time-stratified case-crossover	$PM_{2.5}$, PM_{10}, SO2, NO2, O3, CO interpolated by inverse distance weighting to individual address	China
Kutzinski CAPNETZ air pollution sputum pathogen spectrum community-acquired pneumonia	Air pollution is associated with a distinct pathogen spectrum in community-acquired pneumonia: a prospective multicentre cohort study.	*Pneumonia (Nathan)*	Prospective cohort	City-level $PM_{2.5}$, NO2, SO2, CO up to one month before enrolment	Germany
Zhang Shanxi 11-city time-series pollutant mixtures influenza vaccination	Influenza Incidence in Relation to Air Pollution Mixtures and the Modifying Role of Vaccination in Shanxi Province, China.	*Environ Res*	Multi-city time-series + random-effects meta-analysis	Daily city-mean $PM_{2.5}$, PM_{10}, NO2, SO2, CO, O3 from regulatory monitors	China
Mechanistic toxicology					
Sun Waite review molecular mechanisms respirable silica coal dust asbestos lung disease	Molecular mechanisms of dust-induced lung disease: exploring the role of physicochemical composition and cellular pathways.	*Part Fibre Toxicol*	Narrative review	Respirable crystalline silica, coal mine dust, asbestos; reduced-iron minerals	Global
Occupational & indoor					
Gupta Maharashtra opencast coal mine $PM_{2.5}$ composition oxidative potential	Comprehensive assessment of $PM_{2.5}$ exposures around a regional mining site in India: integrating chemical composition, oxidative potential and health risk	*Environ Geochem Health*	Filter sampling + composition	$PM_{2.5}$ mass, EC/OC, metals, PAHs, OP (AA and DTT) at six sites	India
Kumar firefighter occupational exposure neurotoxicity narrative review	Occupational exposure and neurotoxic health effects in firefighters: toxicological mechanisms and health risks.	*Toxicol Mech Methods*	Narrative review	Fireground combustion products: fine PM, PAHs, CO, HCN	Global
Subramanian chamber size-resolved emission rates PM oxidative potential six indoor sources	Size- and Time-Resolved Emission Rates of Particulate Matter Mass, Chemical Constituents, and Oxidative Potential from Indoor Sources	*Environ Sci Technol*	Chamber emission characterisation	Size-resolved PM mass, composition and DTT oxidative potential, <0.25 to 10 um	USA
Sensing, forecasting & instrumentation					
Agrawal Bihar low-cost sensor network ML exposure inequity short-term mortality	Unveiling hidden air pollution exposure and impacts with low-cost sensor network-based frameworks.	*Nat Commun*	Network deployment + statistical model	$PM_{2.5}$ from a dense statewide low-cost sensor network, gap-filled by machine learning	India
Chen TRACER Houston micropulse lidar aerosol vertical profile boundary layer parameterisation	Aerosol vertical distributions shaped by boundary layer dynamics in a coastal urban environment: insights from the TRACER campaign	*Atmos Chem Phys*	Remote sensing / lidar retrieval	Aerosol concentration profiles from micropulse lidar constrained by surface aerosol measurements	USA
Rogers VUV-AMS siloxane radical mass calibration source aerosol mass spectrometry	Siloxane-Radical Species: Chemical Characteristics and Use for Mass Calibration in Aerosol Mass Spectrometry.	*J Phys Chem A*	Laboratory instrument development	Soot particles carrying vaporised PDMS siloxane oligomers	Chamber / laboratory
Wang $PM_{2.5}$ PM_{10} forecasting dual-data MWS-optimised ensemble learning	$PM_{2.5}$ and PM_{10} Concentration Forecasting Using a Dual-Data Strategy and MWS-Optimized Ensemble Learning	*Atmos Environ*	Metadata only: no abstract in Crossref at issue time	$PM_{2.5}$ and PM_{10}	Not stated

First author	Focus	Journal	Design	Metric	Region
Exposure assessment & modelling					
Ayittey Accra FRM gravimetric $PM_{2.5}$ three sites e-waste 2021-2022	Persistent urban $PM_{2.5}$ pollution in Accra, Ghana: spatiotemporal patterns, meteorological and anthropogenic drivers, and associated health risks	*Environ Monit Assess*	Long-term field measurement campaign	24-h gravimetric $PM_{2.5}$ by USEPA FRM samplers, >300 samples at three sites	Ghana
Hussain Bergin Pakistan smog episodes data-driven source attribution	Attribution of Air Pollution Sources During Smog Episodes Across Major Urban Centers of Pakistan: A Data-Driven Approach	*Atmos Environ*	Metadata only: no abstract in Crossref at issue time	Smog-episode air pollution, likely $PM_{2.5}$-dominated	Pakistan
Jaelani Surabaya Benowo landfill 15-min $PM_{2.5}$ 2020-2024	The spatial impact assessment of landfill sites on air quality in Surabaya: a diurnal and seasonal perspective	*Environ Sci: Atmos*	Statistical analysis of monitoring data	15-minute $PM_{2.5}$ at the Benowo landfill, 2020-2024	Indonesia
Ku North China Plain new particle formation GEOS-Chem-APM spring campaign	Investigations of New Particle Formation Mechanisms in North China Plain during the Spring Campaign Based on Observations and Simulations	*Environ Sci Technol*	Chemical transport model evaluation	Particle number size distributions; nucleation and growth rates	China
Sathish Kanpur Mumbai nine-stage size-resolved aerosol chemistry ICRP deposited dose	Size-resolved measurements matter for chemistry and health: paired inland-coastal size-resolved aerosol chemistry and the inadequacy of bulk PM(2.5) dose estimates.	*Environ Res*	Filter sampling + composition	Nine-stage impactor PM (<0.43 to >9 um), ions and OC/EC; ISORROPIA-II pH and water	India
Schwarz Helmholtz letter ultrafine particle health studies forest for the trees	Letter to the Editor: Health effect studies of ultrafine air pollution-Are we missing the forest for the trees?	*Environ Res*	Letter to the editor	Ultrafine particles	Global
Shankar meteorological drivers urban air quality weather-pollution interactions	Understanding the Meteorological Drivers of Urban Air Quality: Weather-Air Pollution Interactions	*Atmos Environ*	Metadata only: no abstract in Crossref at issue time	Urban air pollutants (unspecified)	Not stated
Tan Shenzhen PMF-ML firework burning $PM_{2.5}$ source apportionment	Apportioning Firework Burning and Routine Source Contributions to PM(2.5) in a Chinese Megacity Using Two PMF-Machine Learning Frameworks.	*Environ Res*	Source-apportionment method	Hourly $PM_{2.5}$ chemical constituents, Shenzhen, Jan-Mar 2023-2025	China
Yao Harbin early-spring heavy pollution episodes natural anthropogenic drivers	Natural and anthropogenic drivers of heavy pollution episodes in early spring in Harbin, Northeast China	*Atmos Pollut Res*	Metadata only: no abstract in Crossref at issue time	Heavy-pollution episodes ($PM_{2.5}$-dominated in Harbin early spring)	China
Burden, policy & mitigation					
Laban scoping review household backup generators emissions health Africa	Health effects associated with emissions from household generators used for backup power in Africa: a scoping review.	*BMC Public Health*	Scoping review	Petrol and diesel generator emissions: $PM_{2.5}$, PM_{10}, NOx, SO2, CO	Sub-Saharan Africa
US coal-fired power plant proximity Area Deprivation Index cross-sectional	Distribution of Coal Power, Pollution, and Poverty.	*JAMA Netw Open*	Population-wide administrative linkage, cross-sectional	Proximity buffers (3, 10, 25, 50 miles) to operational coal-fired power plants; no measured PM	USA
Other clinical endpoints					
Yang US claims case-control $PM_{2.5}$ sinonasal malignant neoplasm	Fine Particulate Matter Exposure and Sinonasal Malignant Neoplasm.	*JAMA Otolaryngol Head Neck Surg*	Case-control (registry)	$PM_{2.5}$ from NASA 1-km satellite-meteorology-land-use models, averaged to residential MSA over 6 months, 1 and 5 years	USA

Provenance and method

Entry window **5 November 2026**, single day (Thursday). Harvester legs: **PubMed** health **12**, sensing **2**; **Europe PMC 0**; **Crossref by ISSN 59**; **arXiv** no in-window entry; **OpenAlex** HTTP 429 on the harvester leg. 72 unique after cross-source dedupe, 0 already carried, **72 fresh**; PubMed connector (EDAT 5 Nov, three axes) 28 PMIDs, 22 bucketed to 5 Nov, 13 appended — **85 screened**; the 6 bucketed to 4 Nov were 3 already carried and 3 off topic. Abstracts: 25 of 26 recovered by efetch/Crossref; the Elsevier day-0 deposits stayed empty. **Amass** BiomedCore (ingest-date filter) 15 returned: 5 already in `seen.json` or rejected, **1 admitted as backfill** (coal-mine composition), the rest previously carried or held. **Consensus** calibration sweep: 10 hits, all matched to the corpus or the rejection log except one (Gökozan, Chemosensors), held for a DOI check. OpenAlex, alphaXiv, bioRxiv/medRxiv and Wiley Scholar Gateway connectors were not available on this session; **Scite** retraction check not run (paid-plan refusal on every prior run), so no record carries a Scite check. **25 admitted** (22 window, 3 backfill), **63 rejected** with logged reasons. **Trial watch** 4–6 November: ClinicalTrials.gov 1 hit (NCT07914010 REACT, carried); Amass TrialCore 15, relevant ones (ACTRN12626001238369, ChiCTR2600134305) already carried — **0 new**. **5 carried on metadata alone**. Affiliations from PubMed XML (OpenAlex unavailable). Abstracts remain the copyright of their publishers.